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A. Mathematical Foundations and Manifold Justification

The SDE in Eq. 3 is $dh(t) = \mu(h(t)) dt + B(h(t)) dW(t)$. Theorem 2.1 states its well-posedness under Lipschitz continuity and linear growth conditions on μ and B . These standard hypotheses guarantee, by classical results (Øksendal, 2003, Thm. 5.2.1), the existence and uniqueness of a strong solution. The proof employs a standard Picard iteration scheme, defining a sequence $(Y^{(n)})_{n \geq 0}$ recursively by

$$Y_t^{(n+1)} = h(0) + \int_0^t \mu(Y_s^{(n)}) ds + \int_0^t B(Y_s^{(n)}) dW_s,$$

$$Y_t^{(0)} = h(0).$$

Standard arguments leveraging Itô isometry (see e.g., Øksendal, 2003) and Grönwall’s lemma (Grönwall, 1919) establish convergence of this sequence to a unique strong solution X_t .

We next address the bound on projection leakage L_k (Definition 2.3). By definition,

$$L_k = \sup_{\substack{x \in \mathbb{R}^D, v^\top V_k = 0, \\ \|v\| \leq \varepsilon}} \frac{\|\mu(x+v) - \mu(x)\|}{\|\mu(x)\|}.$$

Using the Lipschitz continuity of the drift μ (with Lipschitz constant L_μ), for perturbations $\|v\| \leq \varepsilon$:

$$\|\mu(x+v) - \mu(x)\| \leq L_\mu \varepsilon.$$

Assuming that the magnitude of the drift does not vanish on the domain of interest $\mathcal{D}$ (justified empirically), we set $\mu_{\min} := \inf_{x \in \mathcal{D}} \|\mu(x)\| > 0$. This yields the bound:

$$L_k(\varepsilon) \leq \frac{L_\mu \varepsilon}{\mu_{\min}}.$$

We can sharpen this by decomposing $\mu(x)$ into projected and residual components: $\mu(x) = V_k V_k^\top \mu(x) + r_k(x)$, where $r_k(x) = (I - V_k V_k^\top) \mu(x)$ is the residual. Defining the ratio $\rho_k = \sup_{x \in \mathcal{D}} \frac{\|r_k(x)\|}{\|\mu(x)\|}$, the triangle inequality gives a refined bound:

$$L_k \leq \rho_k + \frac{L_\mu \varepsilon}{\mu_{\min}}.$$

Practically, we enforce $L_k \ll 1$ by selecting k large enough to reduce ρ_k (i.e., capture most of the drift direction within a computationally tractable subspace) and restricting perturbations to small ε .

The choice of a rank-40 drift manifold ($k = 40$) is motivated by the impracticality of constructing SDE models directly in the full embedding dimension (e.g., $D \geq 2048$). Empirical PCA on observed drift increments Δh_t (summarized in a data matrix H) shows that the first 40 principal components capture approximately 50% of the drift variance. If $H = U \Sigma W^\top$ is the SVD of H , the relative Frobenius norm of the residual after rank- k truncation is $\sqrt{\sum_{i>k} \sigma_i^2 / \sum_i \sigma_i^2}$. For $k = 40$, this value is $\rho_{40} \approx 0.50$. While this captures only half the variance, it provides a significant simplification that makes the dynamical systems modeling approach feasible. Subsequent components add diminishing amounts of variance. Perturbation theory, specifically the Davis–Kahan sine-theta theorem (Davis & Kahan, 1970), further ensures this empirical drift manifold is stable given the observed spectral gap at the 40th eigenvalue and large sample size. Higher ranks would increase inference complexity with diminishing returns in variance capture for this approximate model, making $k = 40$ a pragmatic choice for balancing model fidelity with the computational feasibility of the SDE approximation. The primary goal is not to claim the random walk *only* occurs on this manifold, but that this manifold serves as a useful and tractable domain for approximation.

Figure 6 shows the distribution of residuals Δh_t projected onto each of these 40 principal component dimensions, revealing rich multimodal structures that motivate the regime-switching approach. These regimes can be interpreted as different reasoning pathways or potential "misaligned states"

that the statistical physics-like approximation aims to capture. While the true multimodality is complex, our four-regime model ($K = 4$) provides an efficient approximation for capturing key dynamics, including deviations that might lead to failures.

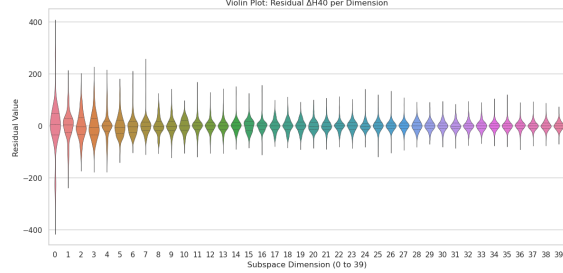


Figure 6. Violin plot of residual Δh_t values projected across the 40 principal component dimensions of the drift manifold (chosen for tractable SDE modeling). Each violin shows the distribution of residuals for a specific dimension, revealing rich multimodal structure that motivates our regime-switching approach. These structures suggest different operational states, some of which could correspond to misaligned reasoning or failure modes.

B. EM Algorithm for SLDS Parameter Estimation

This appendix details the Expectation-Maximization (EM) algorithm (Dempster et al., 1977) used for fitting the parameters of the Switching Linear Dynamical System (SLDS) as defined in Eq. 8. The model parameters are $\theta = (\pi, T, \{M_j, b_j, \Sigma_j\}_{j=1}^K)$, where V_k is a fixed orthonormal PCA projection basis (e.g., $k = 40$, chosen for practical modeling).

The SLDS dynamics are:

$$Z_t \sim \text{Categorical}(\pi) \quad \text{for } t = 0,$$

$$P(Z_{t+1} = j | Z_t = i) = T_{ij} \quad \text{for } t \geq 0,$$

$$h_{t+1} = h_t + V_k(M_{Z_{t+1}}(V_k^\top h_t) + b_{Z_{t+1}}) + \epsilon_{t+1},$$

with residual noise $\epsilon_{t+1} \sim \mathcal{N}(0, \Sigma_{Z_{t+1}})$.

The log-likelihood for observed data $H = (h_0, \dots, h_{T_{\text{end}}})$ is $P(H | \theta) = \sum_Z P(H, Z | \theta)$, where $Z = (Z_0, \dots, Z_{T_{\text{end}}-1})$. Direct maximization is intractable, hence EM. At iteration m , EM alternates:

B.1. E-step

Compute expected sufficient statistics under $\theta^{(m)}$. Use standard forward ($\alpha_t(j) = P(h_0, \dots, h_t, Z_t = j | \theta^{(m)})$) and backward ($\beta_t(j) = P(h_{t+1}, \dots, h_{T_{\text{end}}} | Z_t = j, \theta^{(m)})$) re-

cursions (Rabiner, 1989). Posterior regime probabilities:

$$\begin{aligned}\gamma_t(j) &= P(Z_t = j | H, \theta^{(m)}) \\ &= \frac{\alpha_t(j)\beta_t(j)}{\sum_{i=1}^K \alpha_t(i)\beta_t(i)}, \\ \xi_t(i, j) &= P(Z_t = i, Z_{t+1} = j | H, \theta^{(m)}) \\ &= \frac{\alpha_t(i)T_{ij}^{(m)}\beta_{t+1}(j)}{P(H|\theta^{(m)})} \\ &\quad + \mathcal{N}(\Delta h'_t | M_j^{(m)}x_t + b_j^{(m)}, \Sigma_j^{(m)})\end{aligned}$$

where $\Delta h'_t = V_k^\top(h_{t+1} - h_t)$ and $x_t = V_k^\top h_t$. The $\mathcal{N}(\cdot)$ term is the emission probability of observing h_{t+1} given h_t and $Z_{t+1} = j$. These probabilities help identify transitions between different reasoning states, including potentially misaligned ones.

B.2. M-step

In the M-step, parameters are updated to maximize the expected complete data log-likelihood. The initial state probabilities $\hat{\pi}_j$ are given by $\hat{\pi}_j = \gamma_0(j)$. Transition probabilities $\hat{T}_{ij}$ are calculated as:

$$\hat{T}_{ij} = \frac{\sum_{t=0}^{T_{\text{end}}-2} \xi_t(i, j)}{\sum_{t=0}^{T_{\text{end}}-2} \gamma_t(i)}.$$

The regime-specific dynamics $\{M_j, b_j, \Sigma_j\}$ are determined through a process analogous to weighted linear regression. We define the projected change as $\Delta h'_t = V_k^\top(h_{t+1} - h_t)$ and the projected state as $x_t = V_k^\top h_t$. Augmented regressors $\mathcal{X}_t = [x_t^\top, 1]^\top$ and corresponding augmented parameters $\mathcal{M}_j = [M_j^\top, b_j]^\top$ are utilized. The update for $\hat{\mathcal{M}}_j$ is then computed as:

$$\begin{aligned}\hat{\mathcal{M}}_j &= \left(\sum_{t=0}^{T_{\text{end}}-1} \gamma_{t+1}(j) \mathcal{X}_t \mathcal{X}_t^\top \right)^{-1} \\ &\quad \times \left(\sum_{t=0}^{T_{\text{end}}-1} \gamma_{t+1}(j) \mathcal{X}_t (\Delta h'_t)^\top \right).\end{aligned}$$

From $\hat{\mathcal{M}}_j$, the dynamics matrix $\hat{M}_j$ and bias vector $\hat{b}_j$ are extracted using $\hat{M}_j = \hat{\mathcal{M}}_j(1 : k, :)^\top$ and $\hat{b}_j = \hat{\mathcal{M}}_j(k + 1, :)^top$, respectively. To update the covariance matrix $\hat{\Sigma}_j$, we first define the residuals for each regime j at time t as $e_{jt} = \Delta h'_t - \hat{M}_j x_t - \hat{b}_j$. Then, $\hat{\Sigma}_j$ is computed by:

$$\hat{\Sigma}_j = \frac{\sum_{t=0}^{T_{\text{end}}-1} \gamma_{t+1}(j) e_{jt} e_{jt}^\top}{\sum_{t=0}^{T_{\text{end}}-1} \gamma_{t+1}(j)}.$$

These updates are derived from maximizing the expected complete data log-likelihood.

Scaling techniques are employed during the forward-backward passes to mitigate numerical underflow. When dealing with multiple observation sequences, the necessary statistics are accumulated across all sequences before the parameter updates are performed. Convergence of the Expectation-Maximization algorithm is typically assessed by observing when parameter changes fall below a predefined threshold, when the change in log-likelihood becomes negligible, or when a maximum number of iterations is reached. The inherent property of EM ensuring a monotone increase in the log-likelihood contributes to stable training. Ultimately, the objective is to identify a set of parameters that most accurately describes the observed dynamics of the reasoning process. This includes modeling transitions between different operational regimes, which can be indicative of phenomena such as the onset of failure modes.

C. Adversarial Chain-of-Thought Belief Manipulation

This appendix describes experimental details for the adversarial belief-manipulation results in Section 5.3, focusing on how the SLDS framework can model and predict LLMs slipping into misaligned states, following ICML practice.

C.1. Experimental Design

We studied Llama-2-70B and Gemma-7B-IT under adversarial prompting on twelve misinformation themes (public health, conspiracies, financial myths, AI fears, historical revisionism, pseudoscience, etc.). For each theme/model, paired clean and poisoned CoTs were generated. Clean CoTs used neutral questions (e.g., ‘‘Summarize arguments for and against vaccination’’). Poisoned CoTs interspersed adversarial prompts at predetermined steps to guide the model towards harmful beliefs (misaligned states). Each CoT had ~ 50 sentence-level steps. We collected ~ 100 trajectories per combination, totaling ~ 3000 trajectories. At each step t , we recorded the final-layer residual embedding and a scalar ‘‘belief score’’ from a diagnostic query related to the misinformation. Belief score = $P(\text{True}) / (P(\text{True}) + P(\text{False}))$, where 0 is rejection and 1 is strong affirmation of the false claim (a clear misaligned state).

C.2. Data Preprocessing

Raw hidden-state vectors were standardized (mean-subtracted, variance-normalized per dimension) and projected onto their first 40 principal components (PCA, $\sim 87\%$ variance explained for this dataset, chosen for practical SLDS modeling) using `scikit-learn 1.2.1` (SVD solver, whitening enabled).